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One-Shot Domain Calibration of Mixture-of-Experts Routers

via Gradient Bias Steering

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ARRL: Adaptable, Reliable, and Responsible Learning

Main idea: calibrate the router, not the experts.

One forward/backward pass over 100 domain examples produces a tiny bias that steers borderline Top-K choices, while the frozen MoE and its standard inference path remain intact.

Abstract

Sparse Mixture-of-Experts models are pretrained on broad data mixtures, so their routers may under-use experts that are useful for a narrow deployment domain. We introduce **Domain-Calibrated Router Bias (DCRB)**, a one-shot post-training method for a fully frozen MoE. A 100-example labeled calibration set provides answer-token gradients with respect to router bias; the resulting per-expert importance is normalized, scaled by the layer's own logit dispersion, and folded into the standard gate bias. DCRB improves mathematics, chemistry, and history accuracy while storing only a few kilobytes per domain and adding no custom routing logic at inference.

Motivation and contributions

- Domain knowledge may already exist in the frozen experts, but the general-purpose router may not expose it consistently.
- Full fine-tuning is expensive; dynamic routing rules can add per-token control logic and latency.
- DCRB changes only one scalar per expert per MoE layer and keeps the number of active experts fixed.

Contributions

- Answer-token gradient importance for each router bias.
- Automatic layer-wise scaling from router-logit statistics.
- Subject-level, resource, and routing-behavior evaluation on two Qwen MoE models.

Method: one-shot gradient bias steering

A. DCRB calibration and deployment pipeline

1. **Collect 100 domain examples** — Use a labeled calibration set disjoint from evaluation.
2. **Run one gradient pass** — Teacher forcing; the loss mask includes answer tokens only.
3. **Score routed experts** — Test whether a larger gate bias lowers answer loss.
4. **Normalize and scale** — Center each layer and use $4\,\mathrm{Std}(z_\ell)/N_\ell$.
5. **Write into the gate bias** — Store $\Delta b^{(d)}$ in the standard router linear layer and deploy.

No iterative optimization. Calibration uses no optimizer state, learning-rate schedule, repeated epochs, trainable adapter, or expert-weight update.

B. Gradient score for each routed expert

For active expert i in layer ℓ, differentiating the routed mixture through the selected-expert softmax gives

$$\frac{\partial o_{\ell,t}}{\partial z_{\ell,t,i}} = w_{\ell,t,i}\big(e_{\ell,i}(h_{\ell,t}) - o_{\ell,t}\big).$$

With $g_{\ell,t} = \partial L/\partial o_{\ell,t}$, DCRB accumulates only answer tokens ($m_t = 1$):

$$S_{\ell,i} = -\sum_{t:m_t=1} g_{\ell,t}^{\top} w_{\ell,t,i}\big(e_{\ell,i}(h_{\ell,t}) - o_{\ell,t}\big).$$

Interpretation. Positive $S_{\ell,i}$ means that a small increase of expert i's router bias is locally aligned with lower answer-token loss.

answer-token loss L → MoE output $o_{\ell,t}$ → router logit $z_{\ell,t,i}$ → gate bias $b^r_{\ell,i}$

backpropagation reaches the additive bias directly because $\partial z/\partial b = 1$

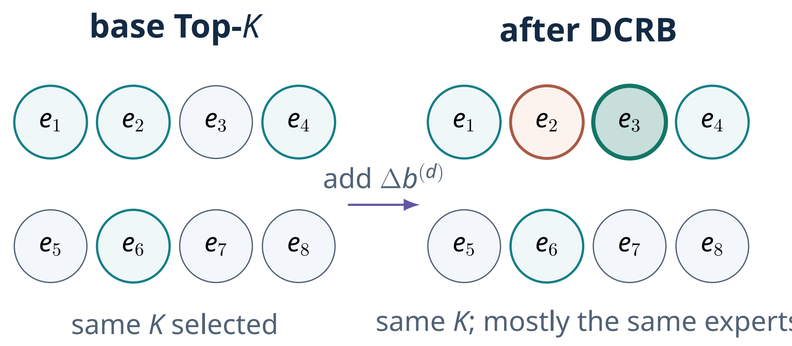
Answer tokens only — Prompt tokens provide context, but only answer-token gradients contribute to $S_{\ell,i}$.

Directly deployable — Because $z_{\ell,t,i}$ depends additively on $b^r_{\ell,i}$, the score targets the same parameter stored at inference.

C. Normalize, scale, and steer only borderline routing decisions

Layer-wise normalization removes score-scale differences and keeps steering conservative within each router:

$$\hat{s}_{\ell,i} = \frac{S_{\ell,i} - \bar{S}_\ell}{\max_j |S_{\ell,j} - \bar{S}_\ell| + \epsilon}, \qquad \Delta b^{(d)}_{\ell,i} = \frac{4\,\mathrm{Std}(z_\ell)}{N_\ell}\hat{s}_{\ell,i}.$$



schematic: **domain expert promoted**, **generic expert suppressed**

Why this scale. $4\,\mathrm{Std}(z_\ell)/N_\ell$ follows each router's natural dynamic range and stays conservative. IQR-, margin-, and squared-score scales were explored as ablations; the 4σ rule was selected for stability.

D. Deployment invariants: a domain adapter measured in kilobytes

- **All pretrained weights frozen** — No expert, attention, embedding, or router matrix is updated.
- **Standard forward path** — The offset is stored as the ordinary bias of the router linear layer.
- **Inference sparsity unchanged** — Top-K and the amount of expert computation remain fixed.
- **One scalar per expert** — Domain variants are tiny bias deltas rather than full checkpoints.

Practical interpretation. DCRB does not teach the model new knowledge or add a runtime module. It changes how a frozen MoE allocates already learned expert capacity to the target domain, while each specialized state remains only a small router-bias vector.

Experiments

1. Evaluation setup

- **Models:** Qwen1.5-MoE-A2.7B (24 layers, 60 routed experts, top-4) and Qwen3-30B-A3B (48 layers, 128 experts, top-8)
- **Calibration:** 100 labeled, non-test examples per domain; teacher forcing; answer-token loss only
- **Evaluation:** MMLU mathematics, chemistry, and history subsets; `all_part` is a held-out 300-example general subset
- **Hardware:** one NVIDIA A800 GPU with 80GB VRAM; all pretrained weights frozen

2. Results at a glance – Qwen1.5-MoE-A2.7B

+5.00 pp College mathematics 35.00 → 40.00

+3.00 pp College chemistry 45.00 → 48.00

+2.45 pp High-school U.S. history 69.12 → 71.57

General behavior is preserved. The held-out `all_part` subset is unchanged after mathematics and chemistry calibration; history calibration changes it by $+0.33$ pp.

3. Mathematics – one fixed DCRB configuration

Bias calibrated on 100 non-test `college_mathematics` examples. Accuracies are percentages.

Evaluation subset	Base	DCRB	△ pp
Qwen1.5-MoE-A2.7B			
College mathematics	35.00	**40.00**	**+5.00**
Abstract algebra	38.00	**39.00**	+1.00
Elementary mathematics	40.21	**41.53**	+1.32
High-school mathematics	30.37	**31.11**	+0.74
`all_part`	55.67	55.67	+0.00
Qwen3-30B-A3B			
College mathematics	73.00	**74.00**	+1.00
Abstract algebra	73.00	**74.00**	+1.00
Elementary mathematics	85.19	85.19	+0.00
High-school mathematics	65.93	**66.30**	+0.37
`all_part`	85.00	85.00	+0.00

Cross-model pattern. Qwen1.5 improves on all four mathematics subsets. The stronger Qwen3 baseline improves or remains unchanged on every mathematics subset and on `all_part`.

4. Chemistry and history – Qwen1.5-MoE-A2.7B

Each bias is calibrated on 100 non-test examples from its named calibration subject.

Evaluation subset	Base	DCRB	△ pp
Chemistry – calibrated on college chemistry			
College chemistry	45.00	**48.00**	**+3.00**
High-school chemistry	44.83	**45.81**	+0.98
`all_part`	55.67	55.67	+0.00
History – calibrated on high-school world history			
High-school world history	70.89	**72.15**	+1.26
High-school European history	68.48	67.88	−0.60
High-school U.S. history	69.12	**71.57**	**+2.45**
`all_part`	55.67	**56.00**	+0.33

Nuance. Chemistry improves on both target subsets. History improves on two of three target subsets; high-school European history decreases by 0.60 pp.

5. Resource accounting and router behavior

<1 min 100 examples on one A800 80GB

2.8KB Qwen1.5: 1,440 FP16 scalars

12.0KB Qwen3: 6,144 FP16 scalars

same Top-K No extra expert computation

Local rerouting. In the most affected Qwen1.5 layers, exact Top-K matches remain around 40%, while partial overlap stays high. Qwen3 reroutes more conservatively.

No collapse. Expert-activation entropy decreases, concentrating mass on a narrower domain-relevant subset, but never falls to zero.

Uncertainty. MMLU subject subsets are small. Changes below one percentage point should not be over-interpreted; the paper emphasizes repeated directionality and preservation of `all_part`, not significance of every individual cell.

Takeaway

A frozen MoE can be specialized by calibrating **expert selection rather than expert weights**. With 100 labeled examples and one gradient pass, DCRB stores only a few kilobytes per domain, improves several target subjects, and preserves the standard Top-K inference path.

Scope. DCRB requires gradient access, cannot add missing knowledge, and is currently evaluated on two Qwen MoE models and MMLU-style multiple-choice tasks.

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